

PORTAGE COUNTY REGIONAL, OH, USA

WMO: 725253

Lat: 41.217N

Lon: 81.250W

Elev: 1198

StdP: 14.07

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00380

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	6.5	-8.0	3.7	3.6	-2.4	5.0	8.7	23.9	27.7	20.6	25.2	5.1	250	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	87.6	70.9	85.1	69.8	82.5	68.3	74.5	81.8	73.0	80.5	71.7	79.2	6.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	125.9	77.4	70.9	119.0	76.2	69.6	113.7	75.4	38.9	81.5	37.5	81.4	36.1	78.9	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.4	13.8	-3.6	91.2	6.1	3.0	-8.0	93.3	-11.6	95.1	-15.0	96.8	-19.4	98.9		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	25.6	28.4	36.9	49.0	60.9	67.5	71.6	69.7	63.6	52.5	40.8	32.7
	DBStd	18.29	11.61	12.15	11.08	9.20	8.54	5.77	5.15	4.80	7.16	8.54	8.91	9.43
	HDD50	2851	757	610	433	126	15	0	0	0	1	70	298	541
	HDD65	6079	1220	1024	871	484	181	40	8	14	112	397	726	1002
	CDD50	2871	2	6	27	97	352	524	668	611	410	149	22	3
	CDD65	624	0	0	0	5	53	113	212	160	71	10	0	0
	CDH74	4869	0	0	8	76	491	876	1696	1136	527	59	0	0
	CDH80	1217	0	0	0	8	92	203	507	259	141	7	0	0
Wind	WSAvg	5.5	7.2	6.8	6.8	7.0	5.1	4.6	3.5	3.3	3.9	5.0	6.1	6.5
Precipitation	PrecAvg	39.2	2.8	2.1	2.8	3.4	4.0	3.5	4.4	3.9	3.6	3.1	2.8	2.8
	PrecMax	51.4	5.9	4.9	5.7	5.7	7.8	5.9	11.4	8.1	7.2	6.1	5.6	6.9
	PrecMin	25.6	1.4	1.0	0.5	1.6	1.6	0.5	1.2	1.1	1.2	1.1	1.4	0.9
	PrecStd	5.9	1.1	1.0	1.2	1.1	1.8	1.5	2.5	1.6	1.6	1.3	1.0	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	65.3	74.2	81.5	86.2	88.4	91.3	88.1	88.3	80.0	70.3	61.0
		MCWB	53.4	54.9	58.8	60.3	68.7	71.0	71.3	71.0	70.0	66.3	57.8	55.6
	2%	DB	54.4	57.4	65.9	75.7	82.3	85.1	87.9	85.6	84.2	74.9	63.5	56.5
		MCWB	51.1	50.8	55.4	58.8	66.0	68.7	71.3	70.4	69.9	63.3	53.4	51.5
	5%	DB	47.7	52.1	60.5	70.8	79.6	82.0	85.5	82.9	80.4	70.8	59.4	52.3
		MCWB	43.3	46.8	51.3	56.2	65.0	67.2	70.6	69.1	68.4	61.4	50.7	47.9
	10%	DB	42.8	45.2	55.0	65.8	76.3	79.1	82.4	80.7	76.3	66.9	55.2	46.8
		MCWB	39.7	39.4	47.0	54.1	62.6	65.5	68.7	68.3	66.5	59.8	48.2	43.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	57.2	61.1	63.5	71.5	73.8	77.1	75.4	75.5	70.1	59.9	57.1
		MCDB	57.1	64.3	69.7	74.3	82.2	83.9	84.4	81.4	81.6	77.1	67.1	59.7
	2%	WB	51.2	53.1	56.8	60.9	69.1	71.8	75.1	73.4	73.0	66.9	56.5	52.9
		MCDB	53.0	55.6	64.1	72.4	77.9	80.7	82.8	80.3	79.7	71.6	60.9	55.3
	5%	WB	44.8	46.1	53.2	58.7	67.2	70.2	73.4	72.1	70.9	63.9	52.4	47.4
		MCDB	46.6	52.4	59.7	68.2	76.4	78.0	80.7	79.5	77.7	68.6	58.4	50.6
	10%	WB	38.3	39.7	48.3	55.7	65.3	68.7	71.9	70.6	68.4	60.5	48.5	42.5
		MCDB	42.3	44.7	54.8	64.2	73.7	75.9	78.9	77.6	74.2	66.3	54.8	46.5
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.5	16.8	20.3	20.3	19.0	19.2	19.2	19.2	17.4	15.2	11.7
		MCDBR	19.0	21.9	23.8	27.0	23.8	22.3	21.6	22.2	22.8	23.0	22.6	18.7
		MCWBR	16.6	16.5	14.6	14.8	11.3	10.6	9.0	9.6	10.2	12.8	15.2	16.6
	5% WB	MCDBR	16.7	20.5	22.0	23.5	20.9	19.0	17.7	17.6	18.9	18.8	19.3	18.0
		MCWBR	16.2	17.0	15.0	15.1	10.6	10.4	8.7	8.6	9.5	12.2	15.9	16.9
Clear-Sky Solar Irradiance	taub	0.299	0.327	0.355	0.386	0.424	0.445	0.448	0.436	0.383	0.359	0.317	0.300	
	taud	2.433	2.329	2.335	2.292	2.206	2.187	2.195	2.222	2.345	2.410	2.504	2.480	
	Ebn at Noon	272	278	283	281	271	265	263	262	270	264	264	262	
	Edh at Noon	25	32	36	40	45	46	45	43	35	30	23	22	
All-Sky Solar Radiation	RadAvg	444	724	1033	1412	1690	1859	1863	1666	1335	860	553	362	
	RadStd	40	90	105	135	146	146	129	84	125	79	84	35	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	N/A

Nomenclature: See separate page